

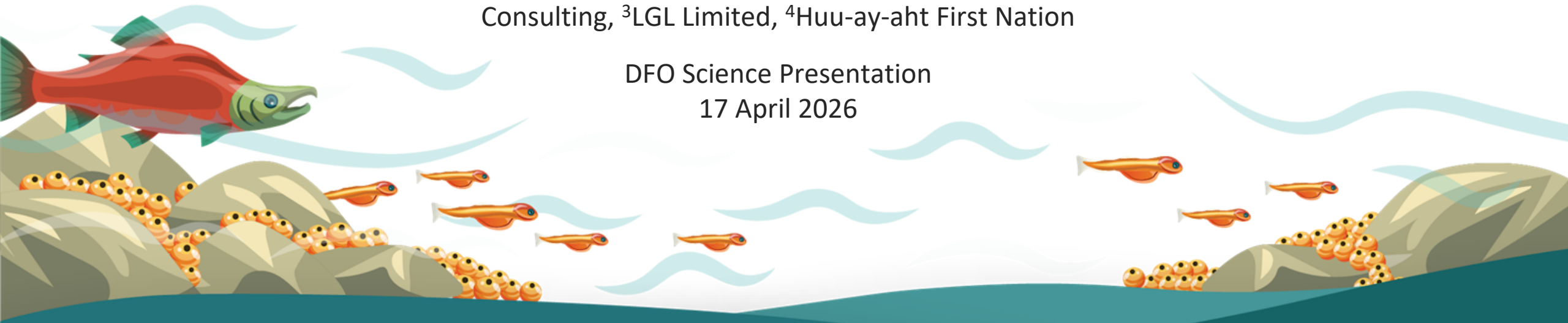
salmonMSE

A decision-support tool to evaluate harvest, hatchery and habitat management decisions for Pacific salmon

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Consulting, ³LGL Limited, ⁴Huu-ay-aht First Nation

DFO Science Presentation
17 April 2026





Technical Advisory Group (28 members)

DFO Core Science
DFO Area Stock Assessment Staff
DFO Fisheries Management
DFO Salmonid Enhancement Program
DFO SARP
First Nations- Island Marine Aquatic,
Nuu-chah-nulth Tribal Council, Ha'oom
Fisheries Society



Sarita Chinook Working Group (11 members)

Huu-Ay-Aht First Nation
DFO Salmonid Enhancement Program
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DFO Core Science

Acknowledgements

Blue Matter Fisheries Consulting, Huu-ah-ayt First Nation, LGL Limited, Fuse Consulting (graphic design)



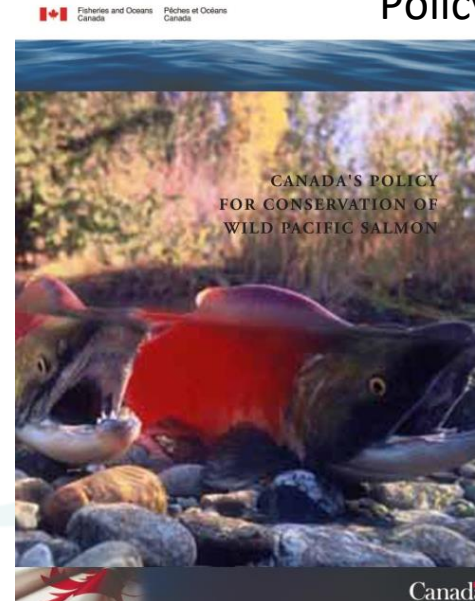
Fisheries Act



Hatchery modernization



Wild Salmon Policy

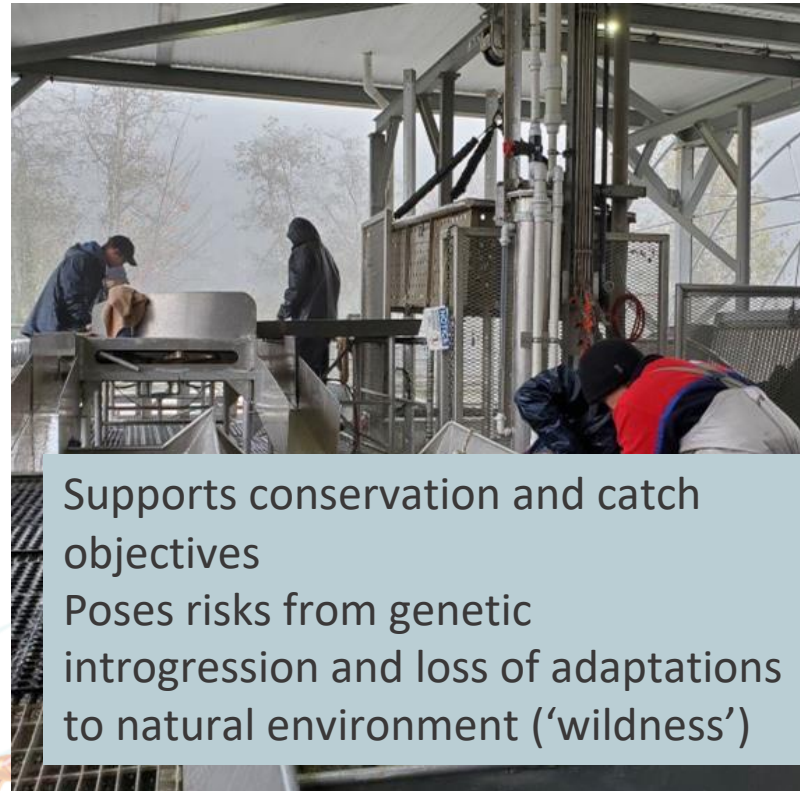


Species at Risk Act



DFO's Reconciliation Strategy: Treaties and Reconciliation Agreements





Photos courtesy of DFO

salmonMSE

this tool can provide a risk-based approach for prioritizing management levers



identify trade-offs in achieving biological, hatchery and harvest objectives for medium-to-long term strategic planning

Generalizable tool, applicable across salmon species and populations

Builds off previous simulation tools:
samSim, AHA, openMSE:

- accounts for risk and uncertainty
- life-stage based (freshwater and marine) specific to Pacific salmon
- includes hatchery demographics, PNI
- modular/flexible to allow for different data availabilities and limitations
- documented and open source



Mark Hobson 2024

Example questions salmonMSE can address

How do mark-selective fisheries impact harvest and conservation objectives?

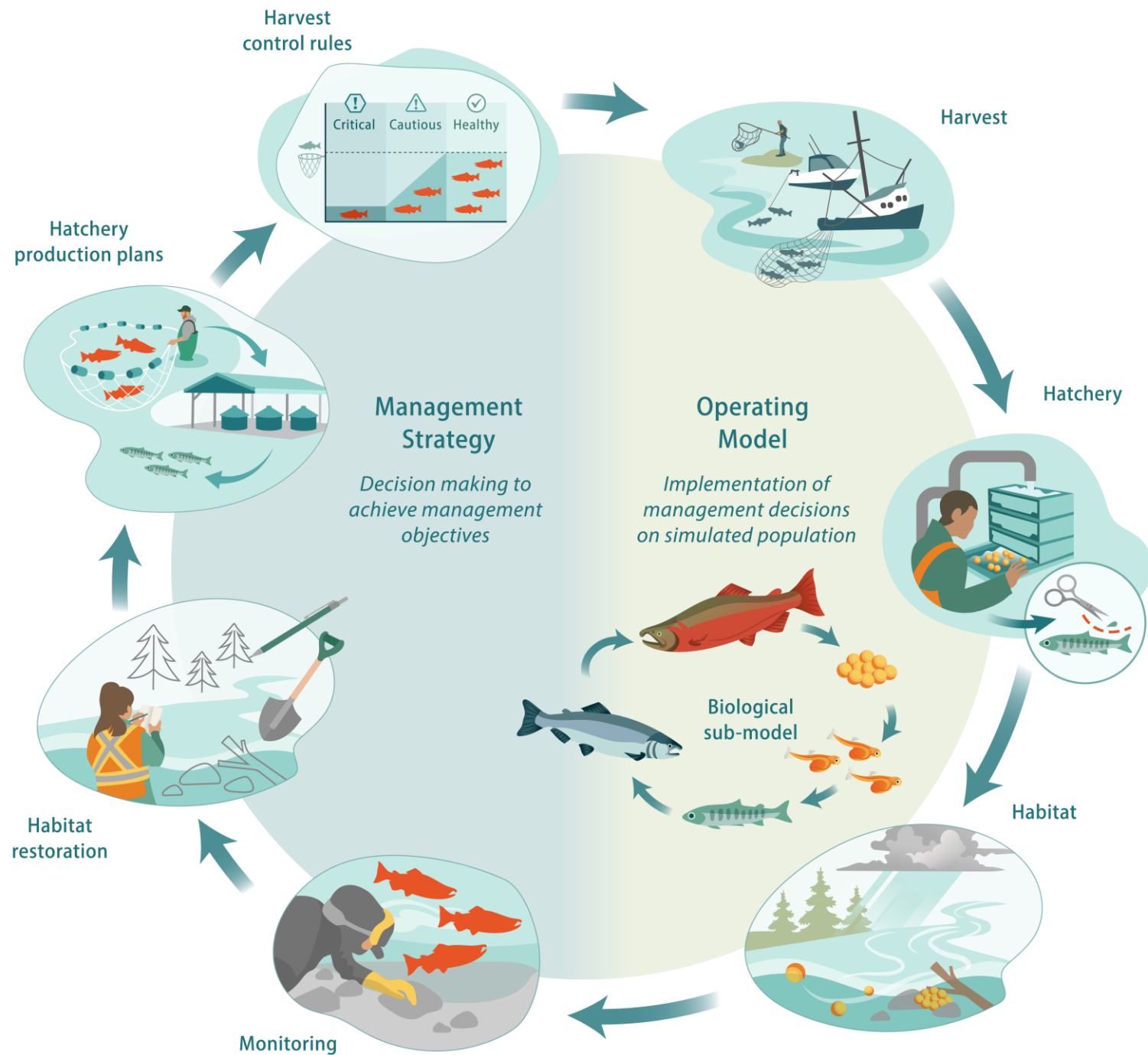
How does a change in hatchery release strategy impact harvest and conservation objectives?

How does the broodtake rule impact the ability to achieve PNI and conservation objectives?

How do changes in environmental drivers impact ability to rebuild?

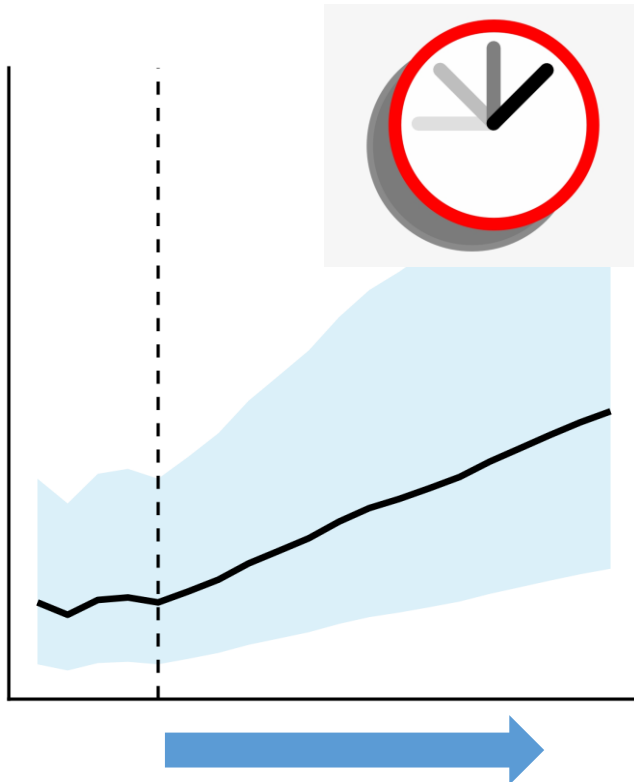


A calculator for analysts and decision makers to evaluate and compare impacts of various types of management levers simultaneously





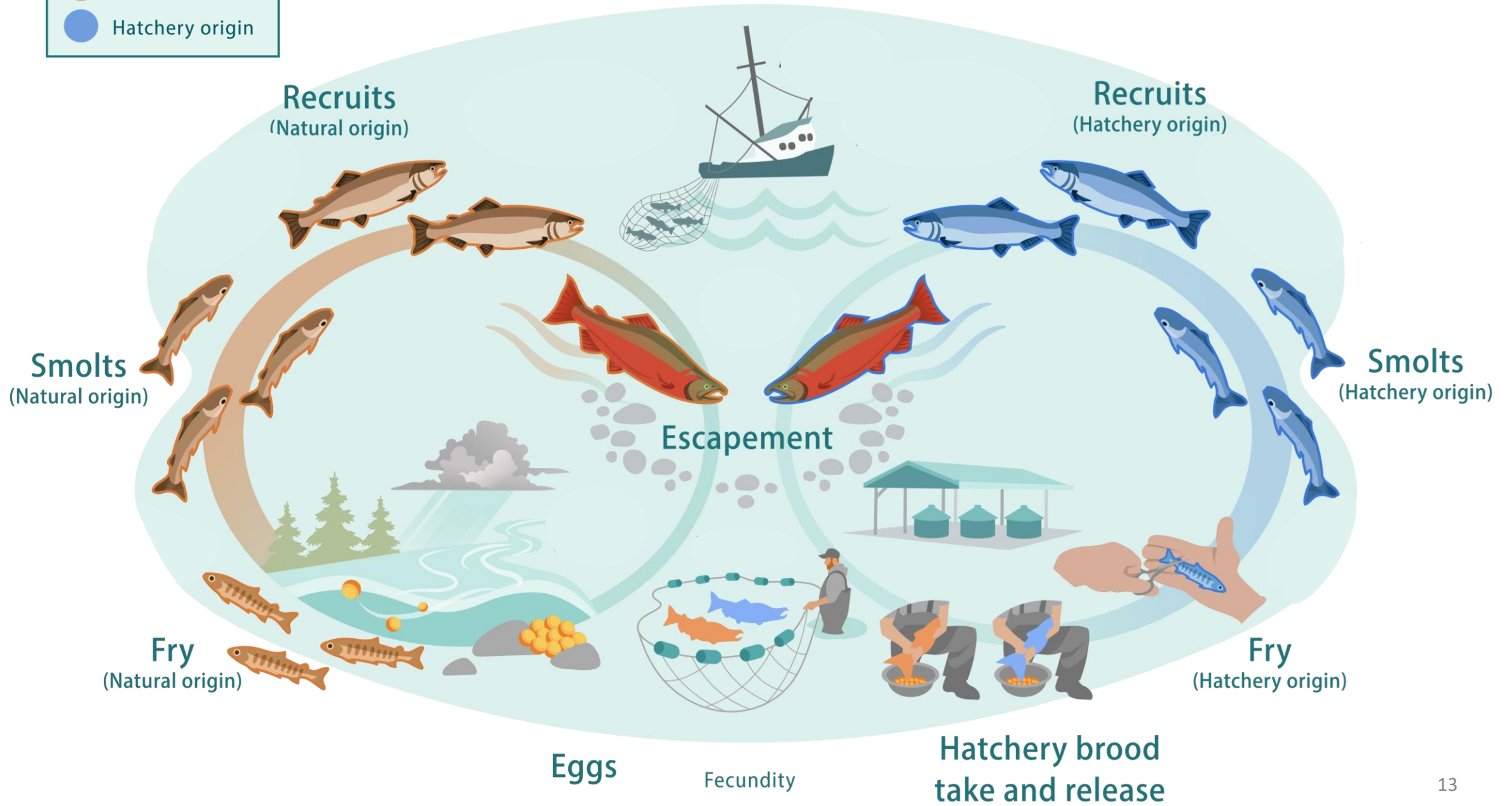
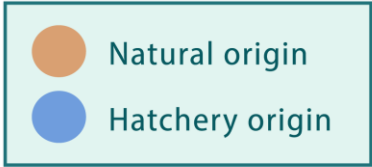
Stochastic forward projections



Evaluate risks

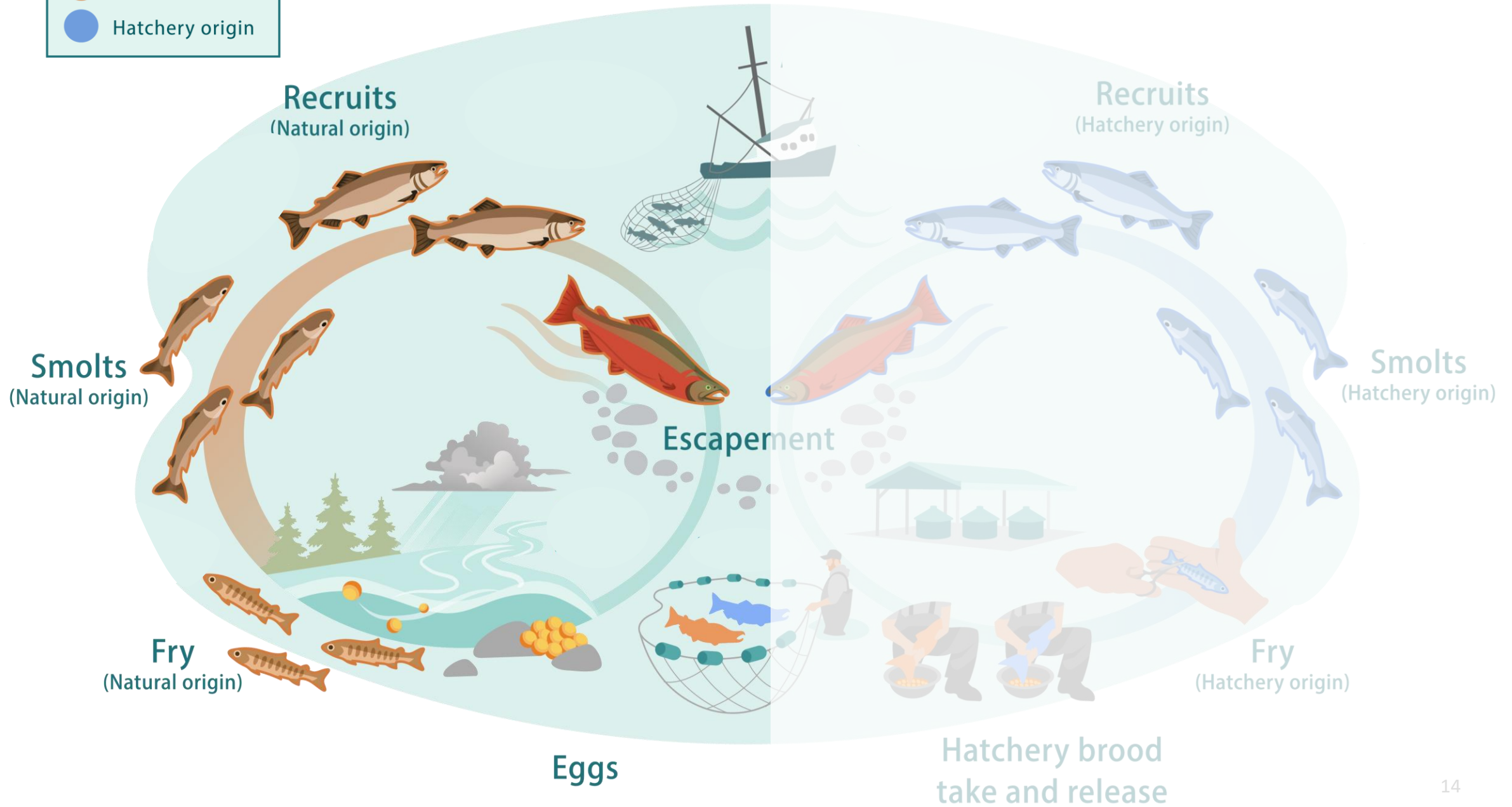


Related to achieving:
Conservation/rebuilding objectives
Catch objectives
'Wildness' objectives



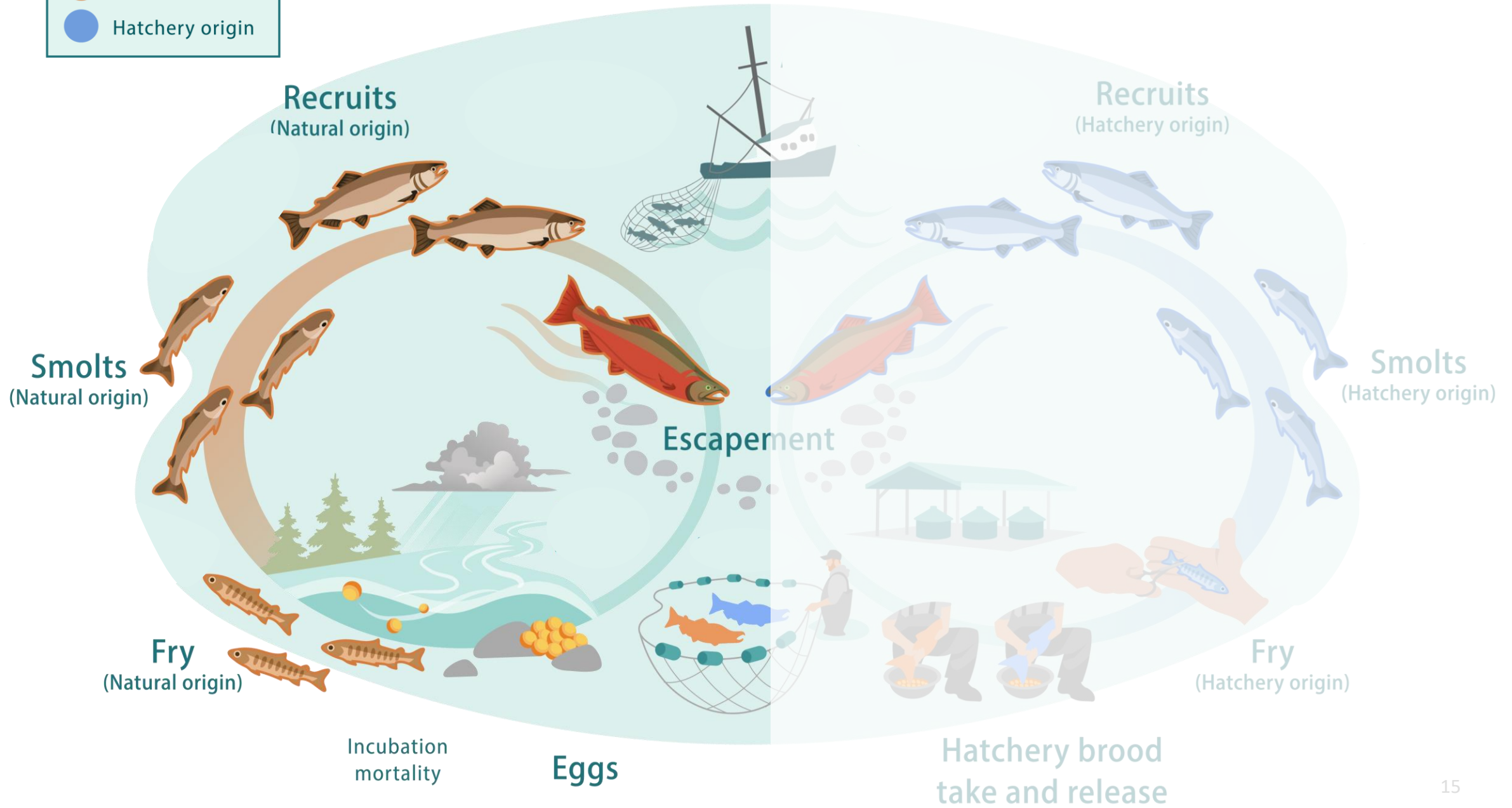
Natural origin

Hatchery origin



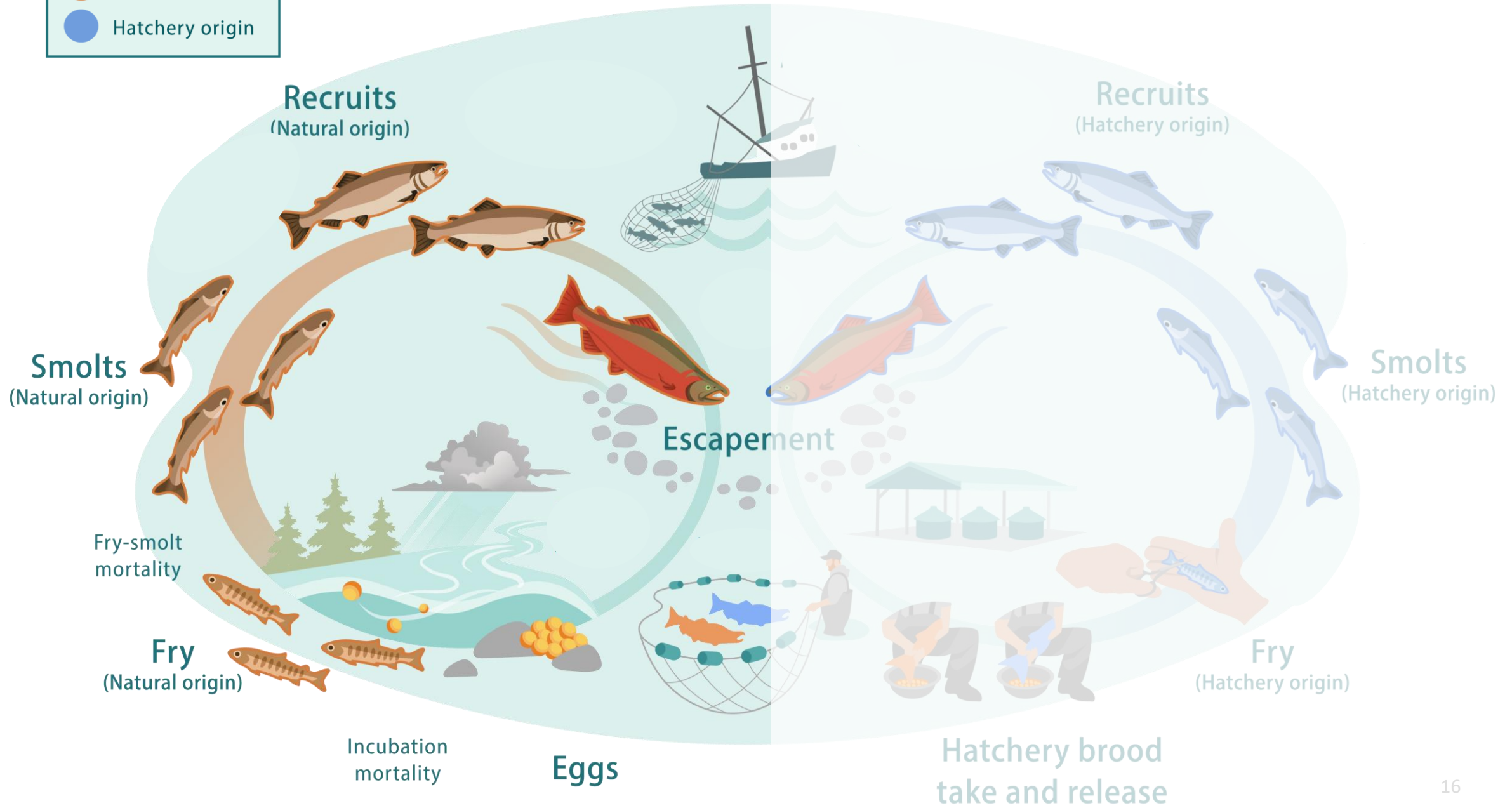
Natural origin

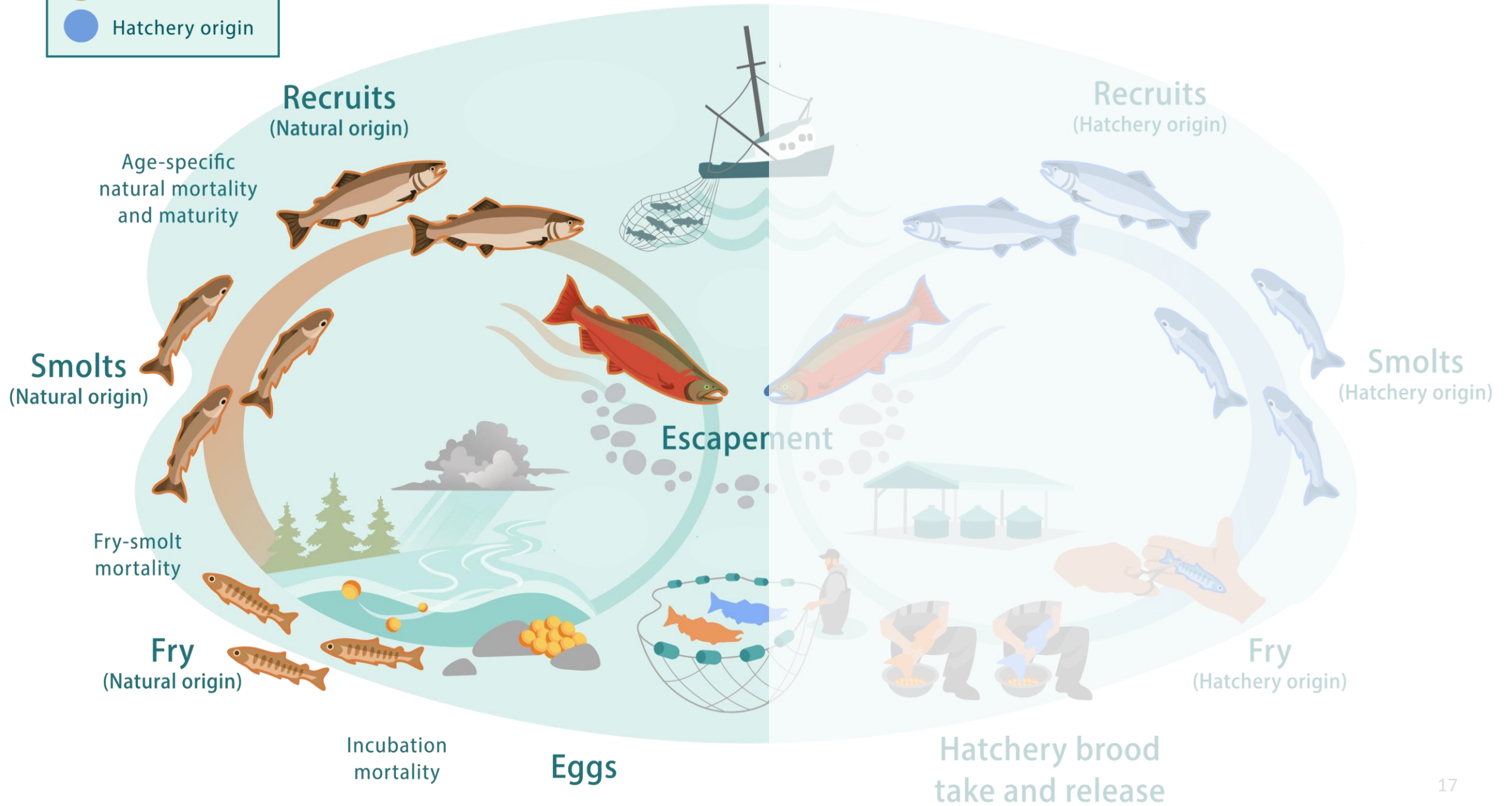
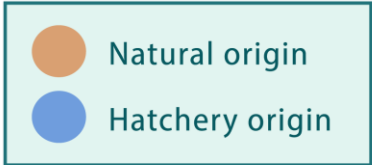
Hatchery origin

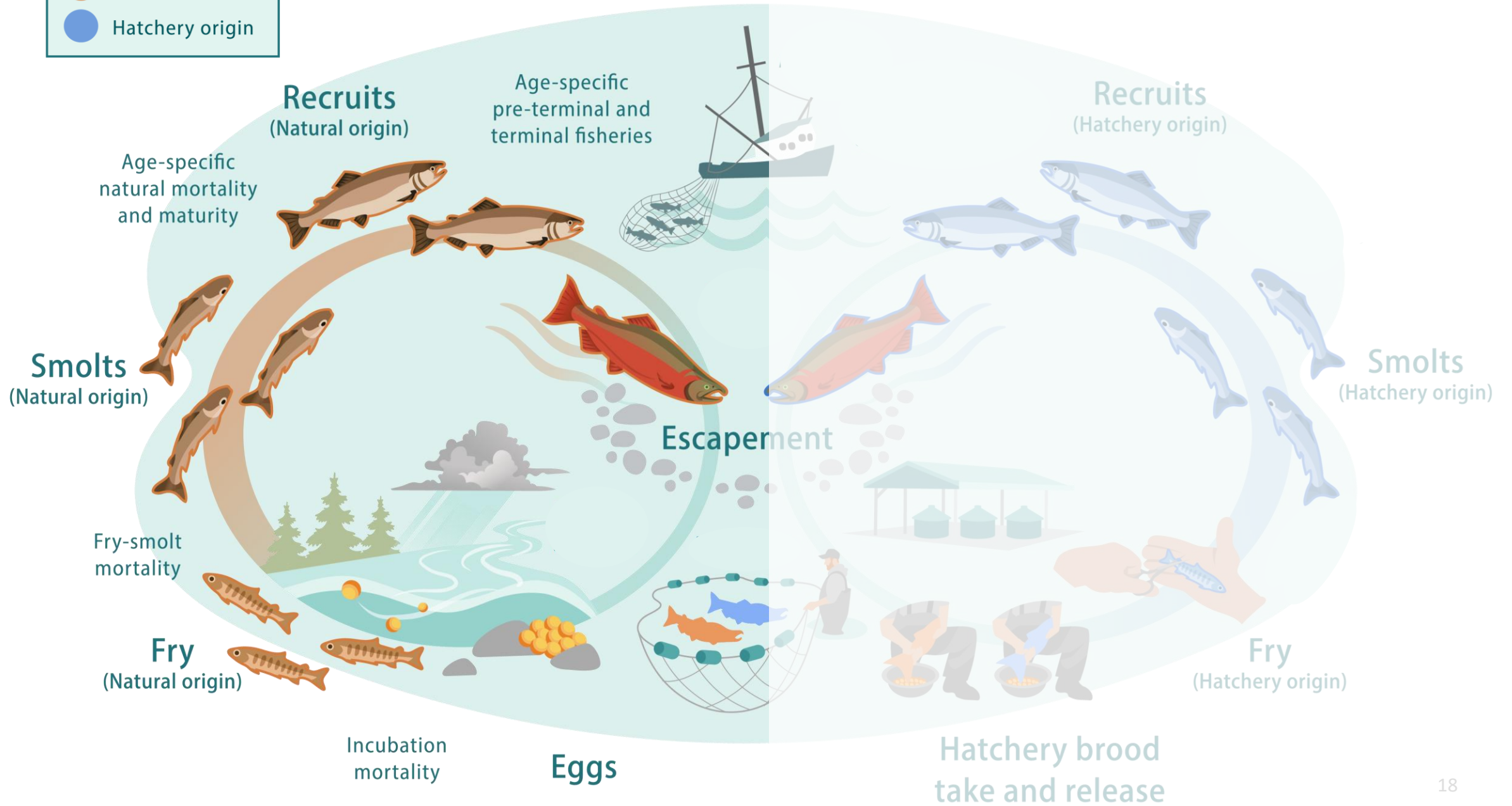
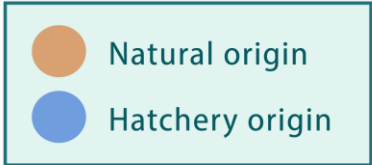


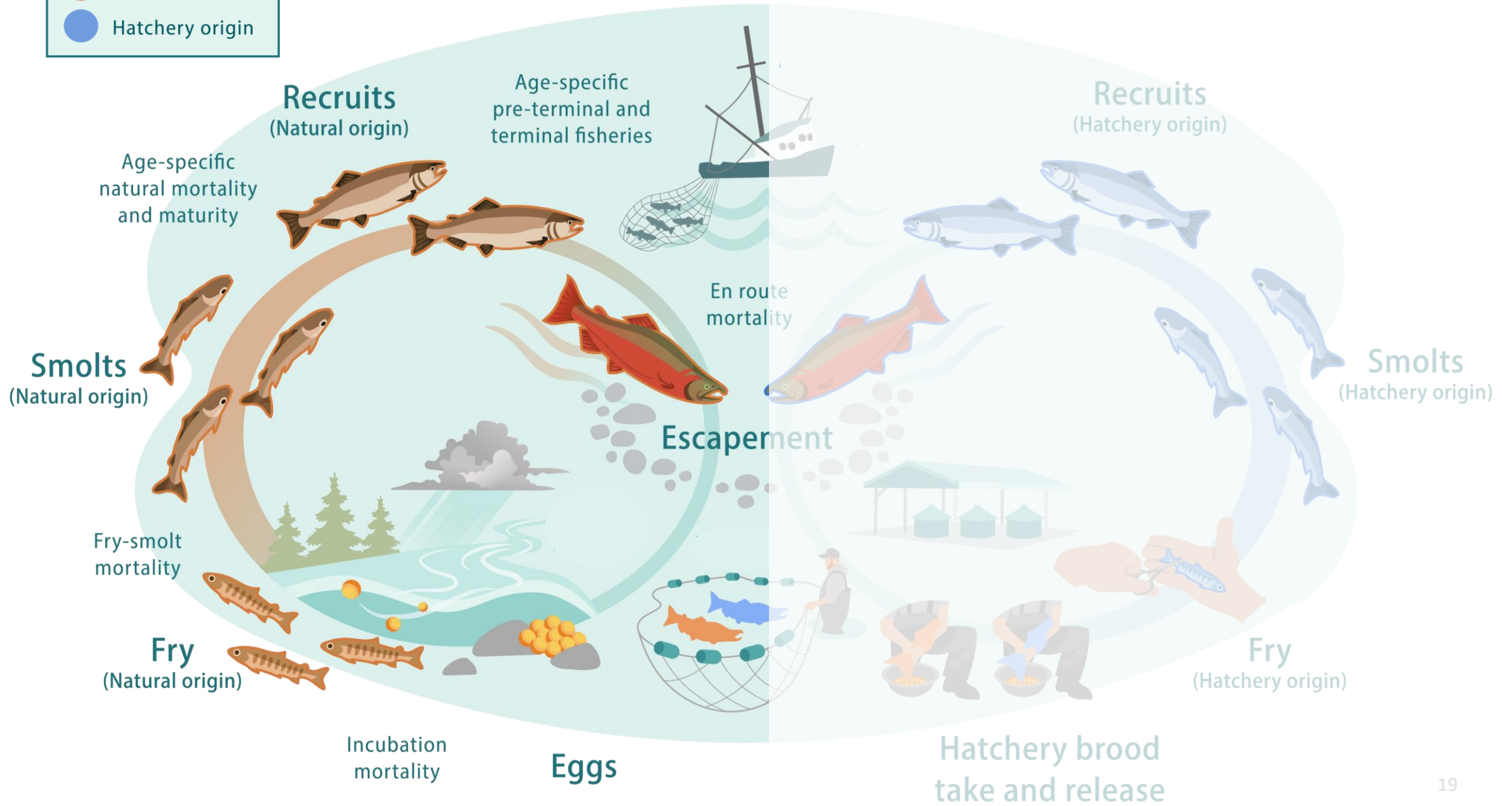
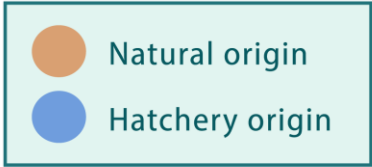
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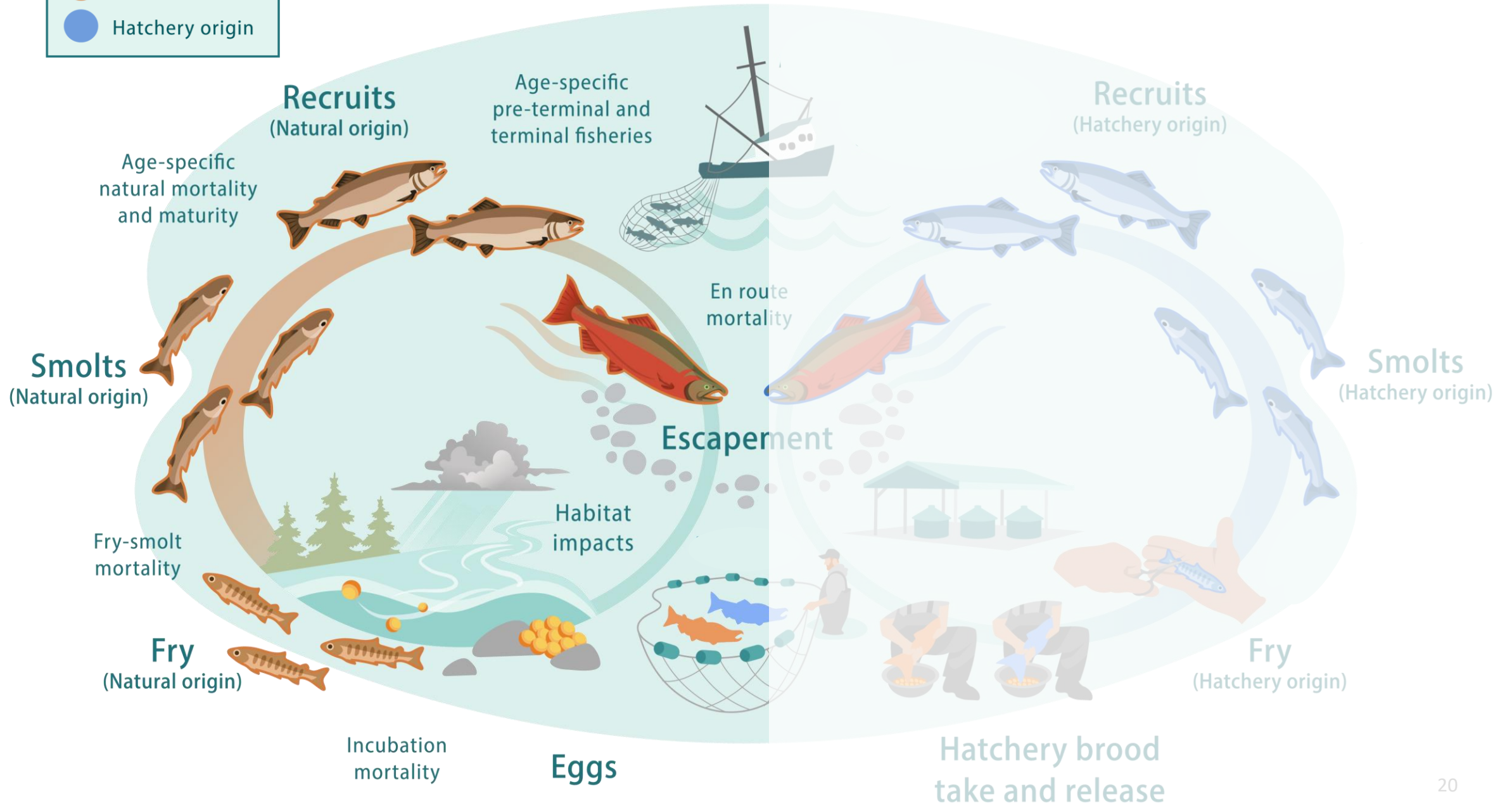
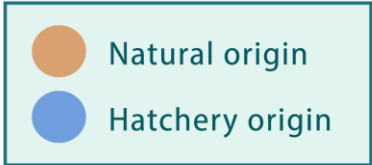
Hatchery origin

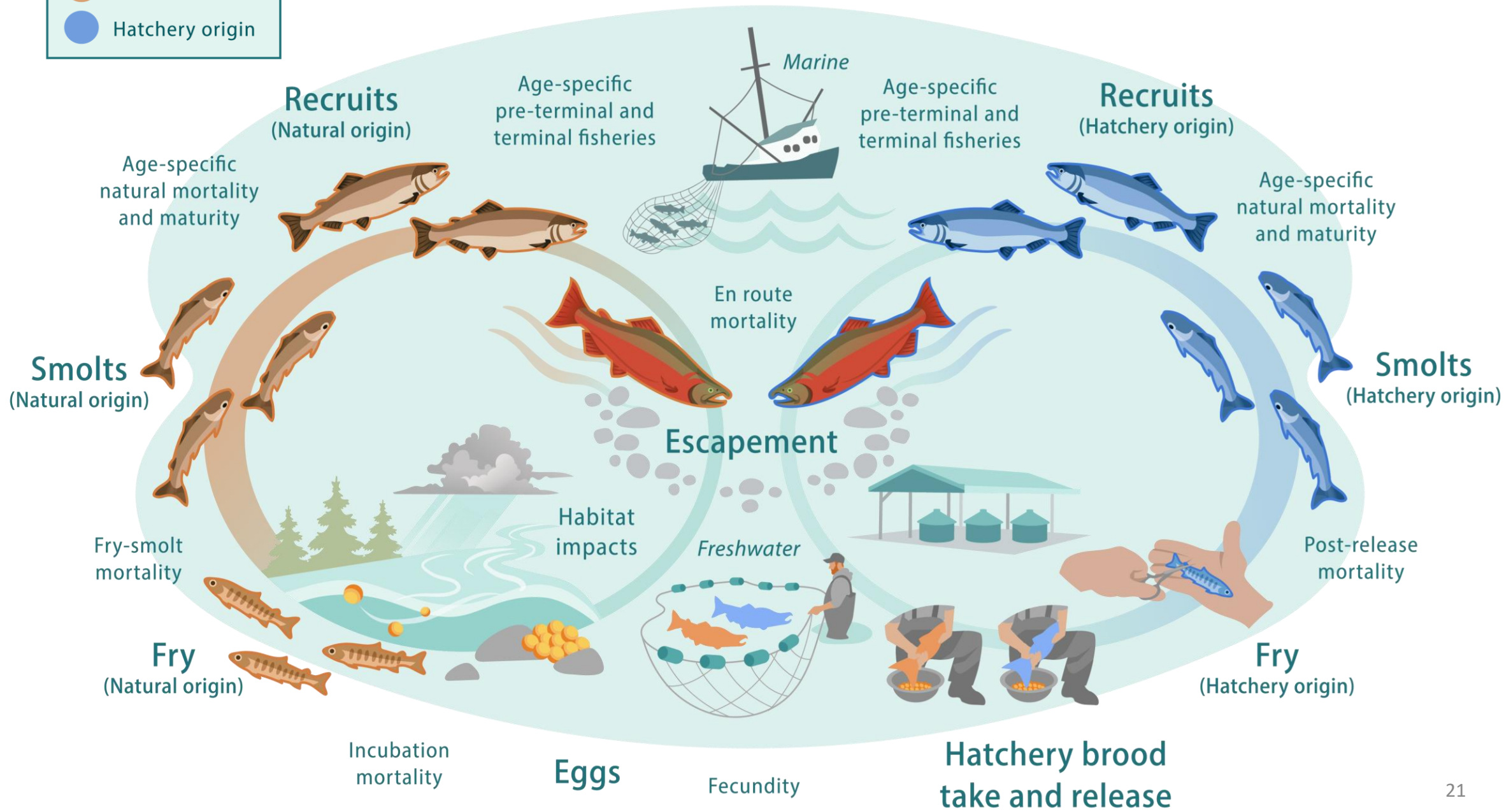
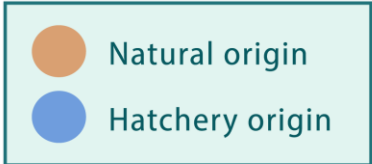




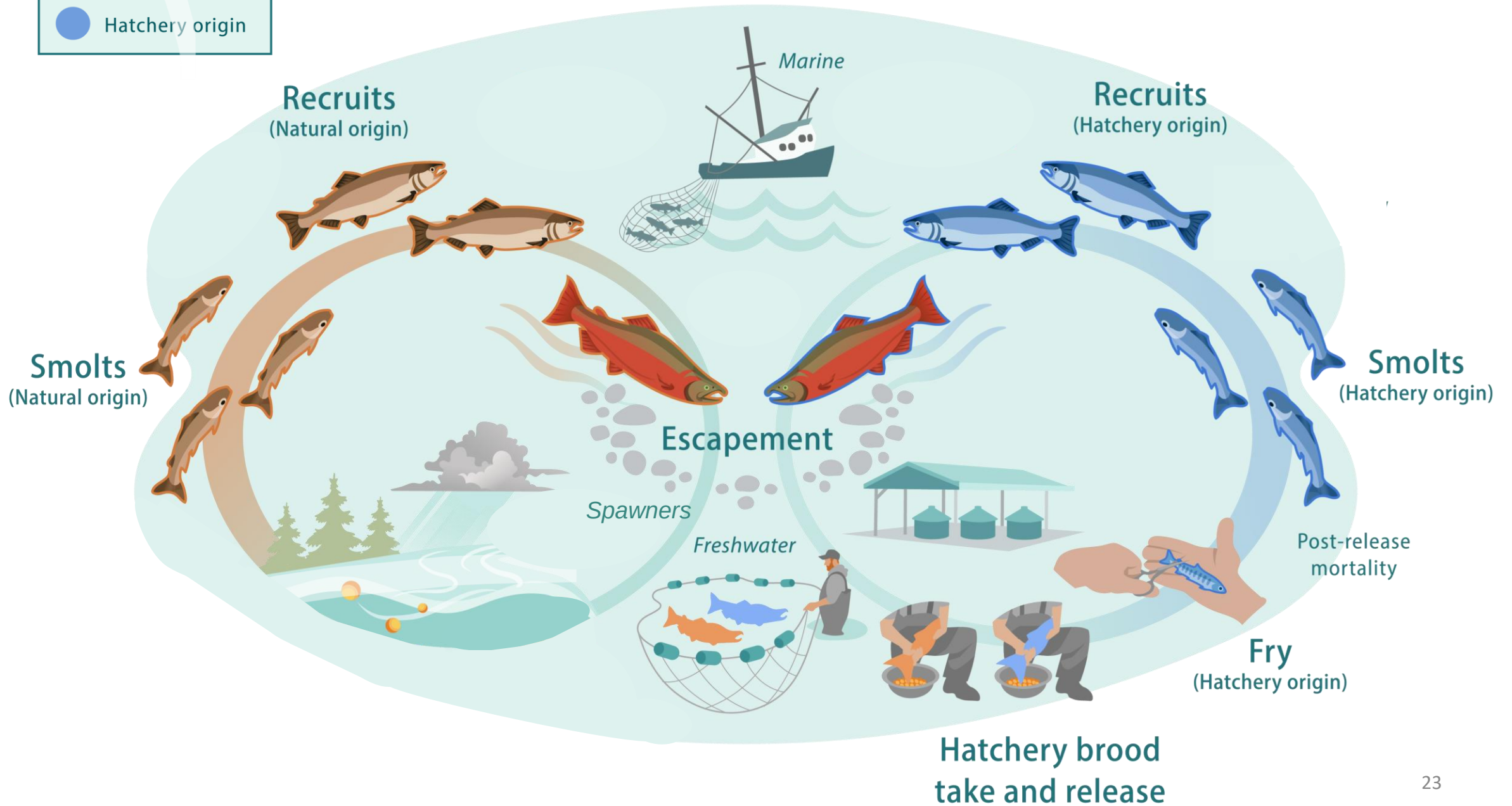
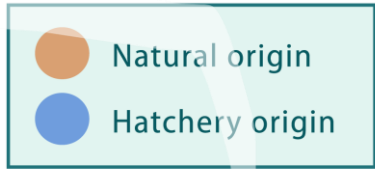








Flexibility for gradient in data availability



Harvest levers

- Pre-terminal, terminal and/or in-river fisheries
- Mark-selective fisheries
- Harvest control rule depending on abundances
 - E.g., minimum conservation threshold before opening in-river fisheries

Hatchery levers

- Number of release groups (0-2), option for yearling or sub-yearling
- Target number released (per release group)
- Mark rate
- pNOB target: Target proportion of the natural-origin brood
- Minimum escapement before brood can be taken
- Maximum proportion of total escapement (after en-route mortality) to use as broodtake (e.g., 33%)
- Maximum proportion of natural-origin escapement to use for natural-origin brood (e.g., 50%)

Performance metrics

- ✓ Probability of meeting **conservation** or rebuilding targets
- ✓ Degree of '**wildness**' of the population (PNI, Proportionate Natural Influence)
- ✓ **Catch** and variability in catch



R package

Our goal is for salmonMSE to be broadly used across DFO, promoting consistency in provision and visualization of results for FSARs, Rebuilding plans, and SEP production plans

www.salmonMSE.com

Sarita River Chinook



Photo credit:
Huu-ay-aht and
Kyler Vos.

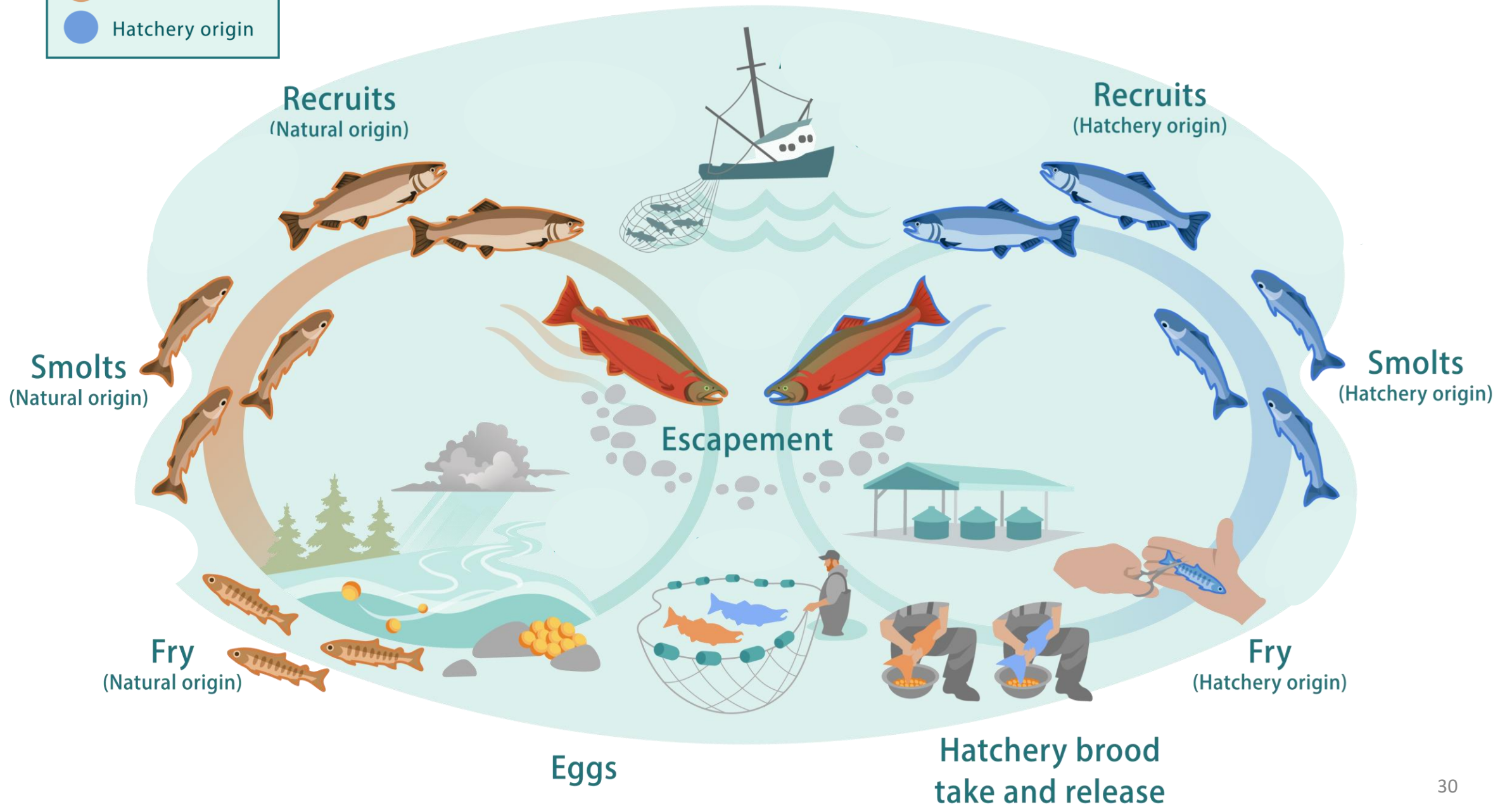
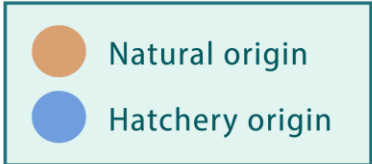


Sarita River Chinook

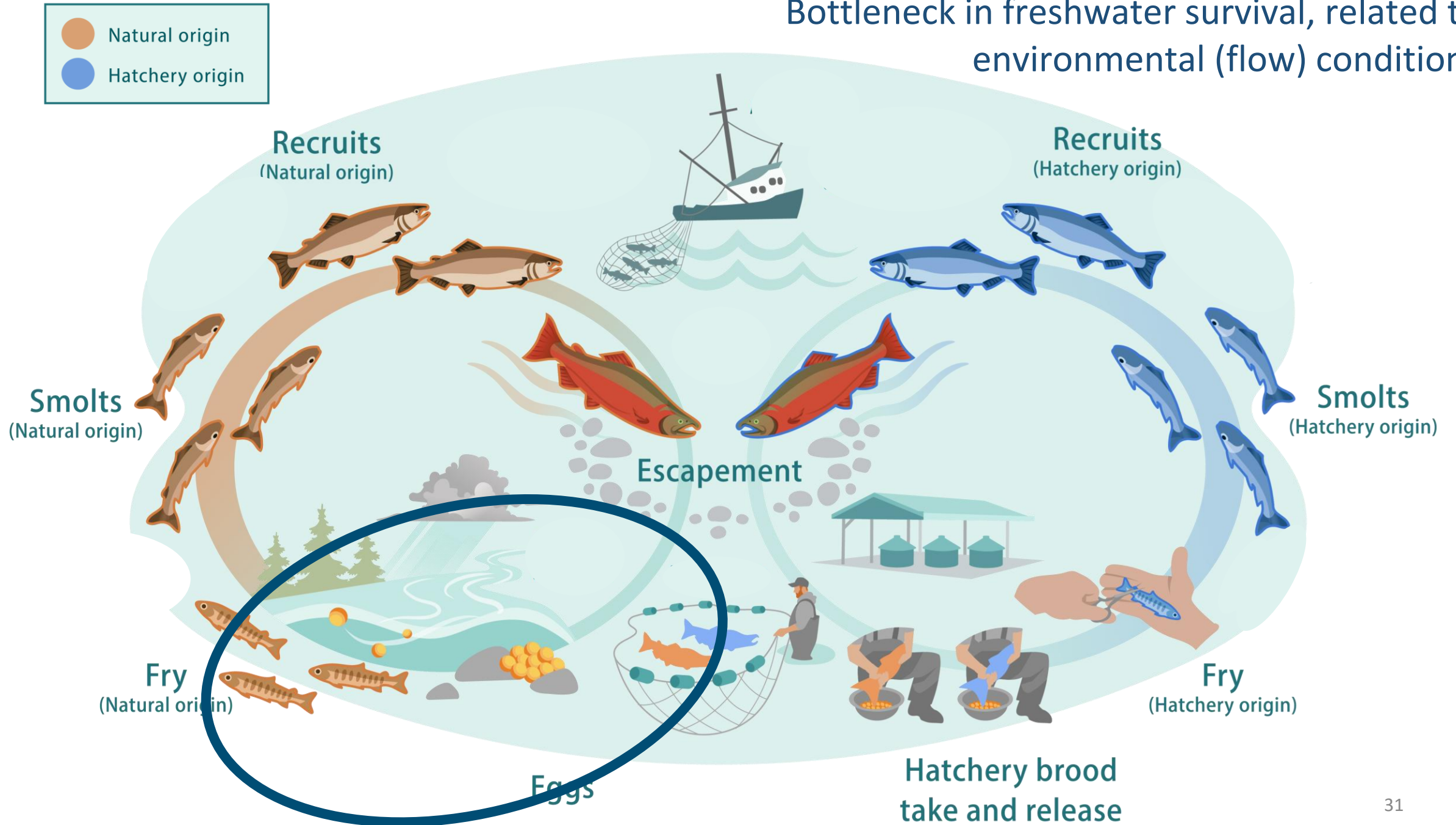


Sarita Chinook Working Group
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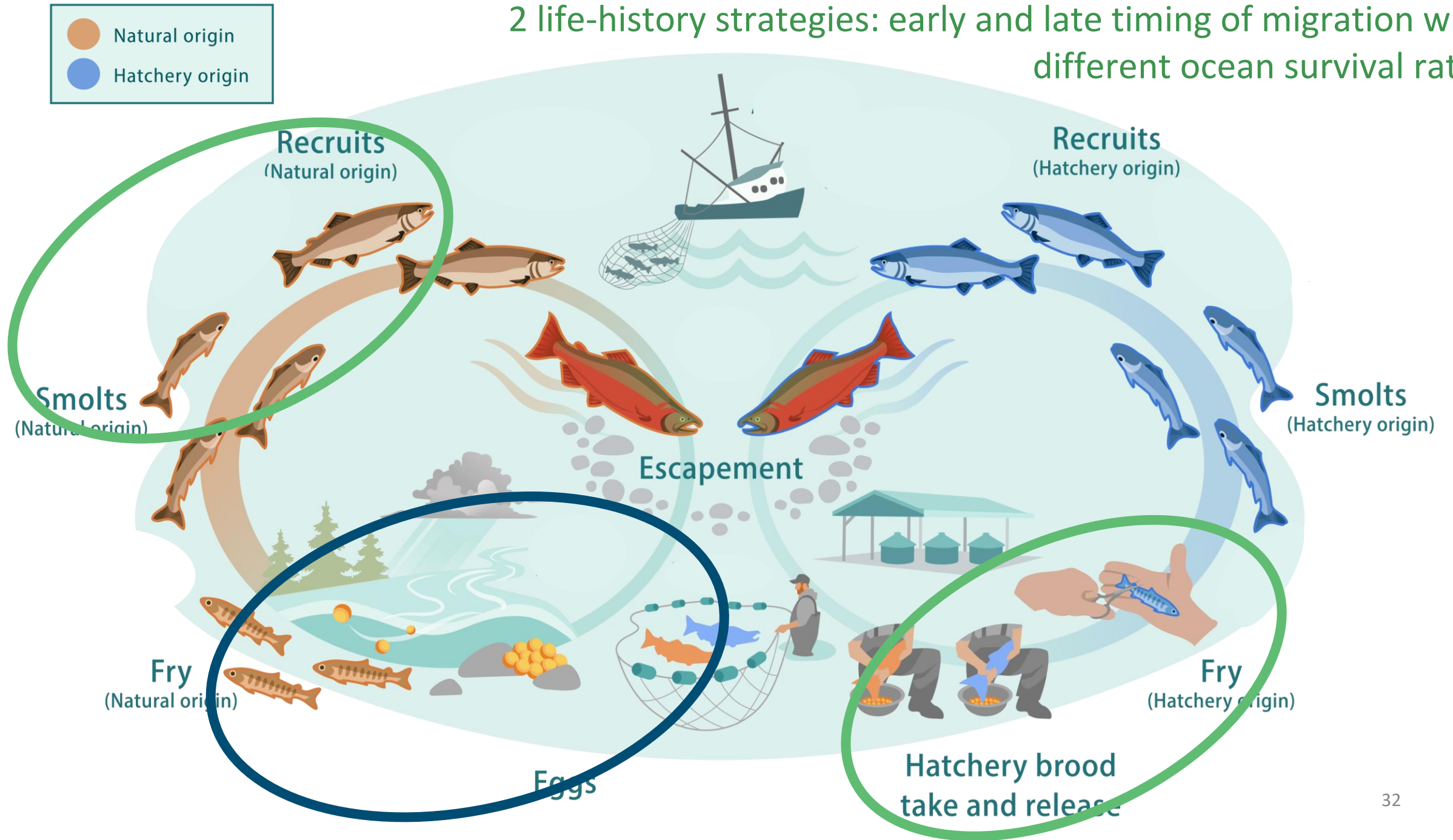




Bottleneck in freshwater survival, related to environmental (flow) conditions



2 life-history strategies: early and late timing of migration with different ocean survival rates



(1) Can the population rebuild under range of harvest and hatchery management options?

(2) Can the population maintain 'wildness' under a range of harvest and hatchery management options?

How do freshwater flow conditions impact our ability to achieve objectives?
How do observed declines in age-at-maturity and fecundity affect these?

Proposed management levers

- In-river harvest management
- Hatchery: Target **P**roportion of **n**atural-**o**origin fish used as **b**rood (**pNOB target**)
- Freshwater habitat restoration

Candidate performance metrics for Sarita River Chinook

- Rebuild and maintain population above 1300
- Maintain 'wildness', with PNI >0.5
- In-river exploitation to support Huu-ay-aht fisheries

Parameterizing projections

Estimation from escapement, hatchery releases, and CWTs

Sarita River 2015-2021 average (based on Laforge et al. 2024)

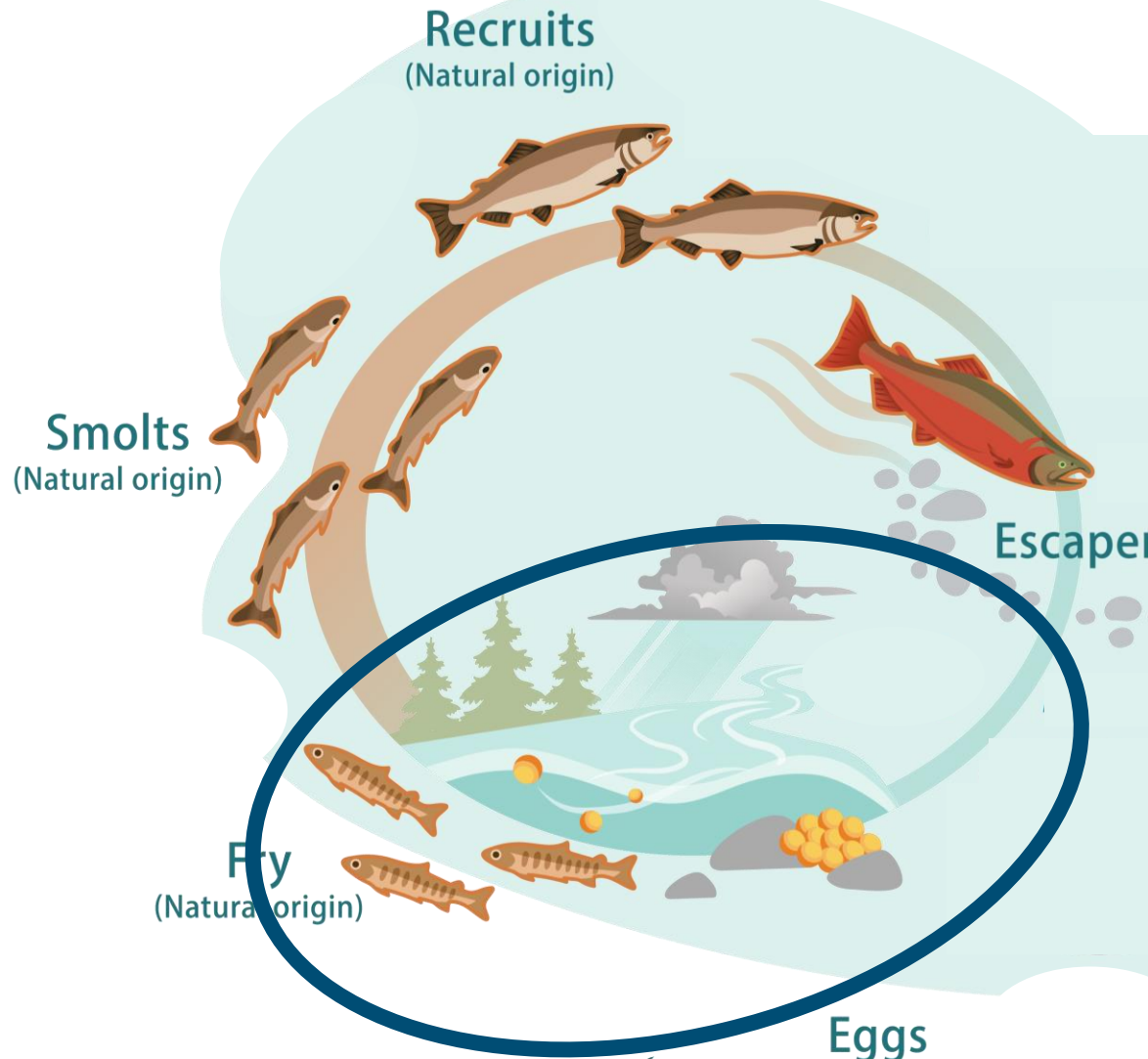
Ecosystem Units	Life Cycle Stage	Number natural origin Chinook	Average Mortality
Estuary / River	1. Adult return to estuary - river mouth	2500	10.0%
River	2. Adult spawners (avg fecundity 3900, 40% female)	2,047	
River	3. Eggs Incubating. <i>Incubation mortality</i>	3,193,881	89.2%
River	4. Inriver post-hatch fry, natural mortality.		20.0%
River / Estuary	5. Fry/Smolts out. <i>Total egg-smolt mortality</i>	274,713	91.4%
WCVI marine	6. Juvenile first marine year natural mortality.		99.5%
CBC marine	7. Juvenile second marine year natural mortality.		30.0%
NBC marine	8. Age 2 abundance. <i>Total smolt to age2 mortality</i>	1,058	99.6%
North Pacific	9. Ages 3-6 rearing abundance after natural mortality.	635	40.0%
Coastwide	10. Preterminal fisheries mortality.		35.0%
WCVI marine	11. Return migration natural mortality along WCVI.		10.0%
WCVI marine	12. Terminal marine fisheries mortality.		5.0%
WCVI marine	13. Adult return. <i>Total mortality smolt to adult return</i>	353	99.9%
Estuary and in-river	14. Add non-local strays		
Estuary and in-river	15. Estuary / in-river fisheries mortality.	-2	
Estuary and in-river	1. Adult return to estuary - river mouth / pre-spawn	351	

Expanded to include information from semi-quantitative life-cycle models derived from meta-analyses, neighboring stocks, and expert opinion that provide **finer resolution, life-stage specific survival values**



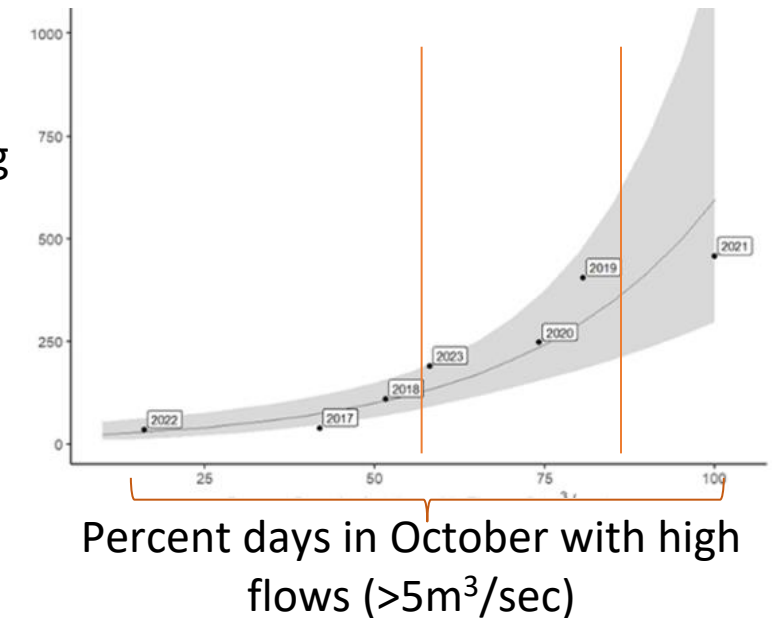
Parameterizing projections

Including environmental covariates



Fry /returning
adult

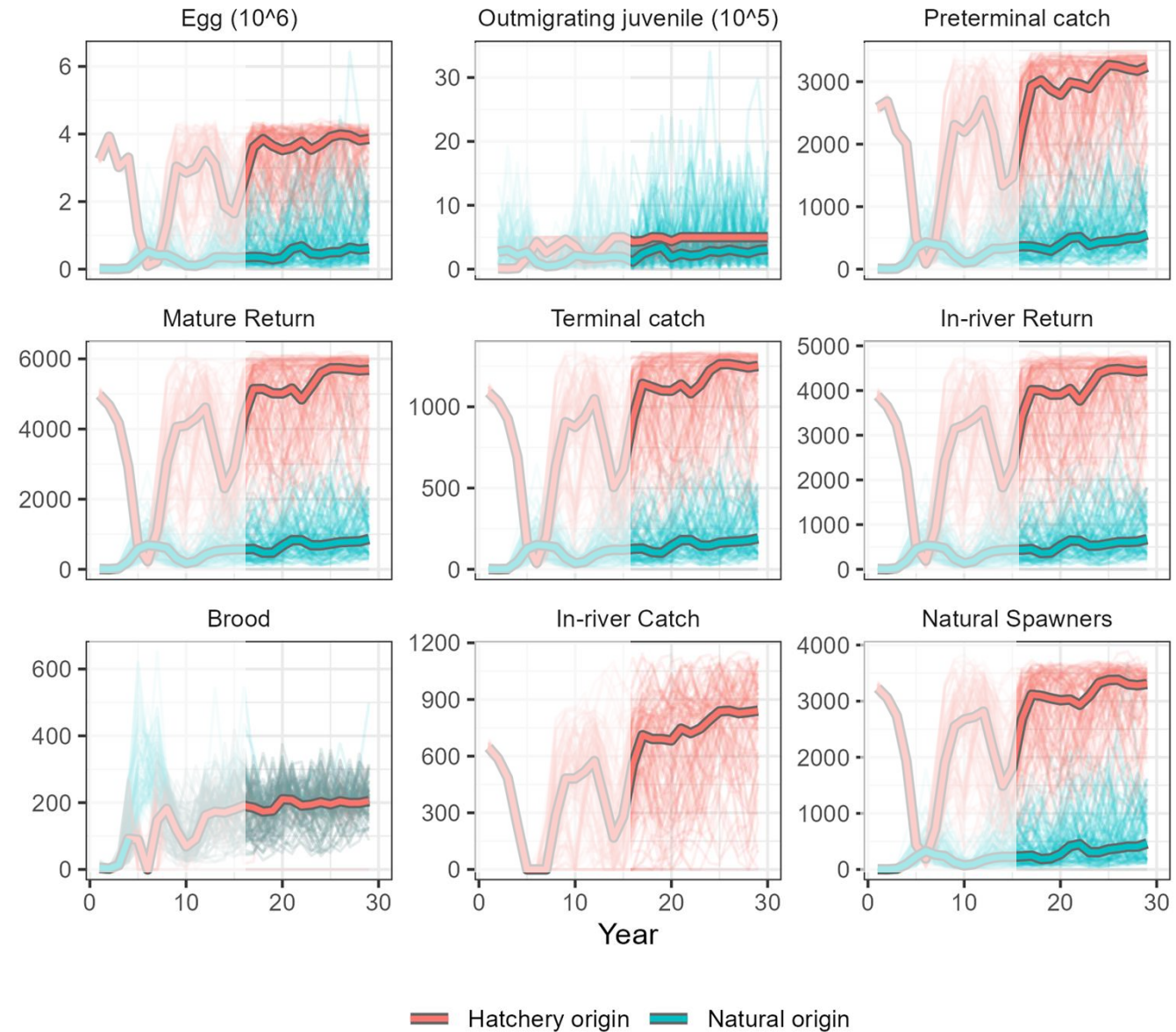
Field studies Sarita River 2017-2023



Run sensitivity analyses corresponding to hypotheses about environmental conditions & habitat restoration that can alter how flows impact survival

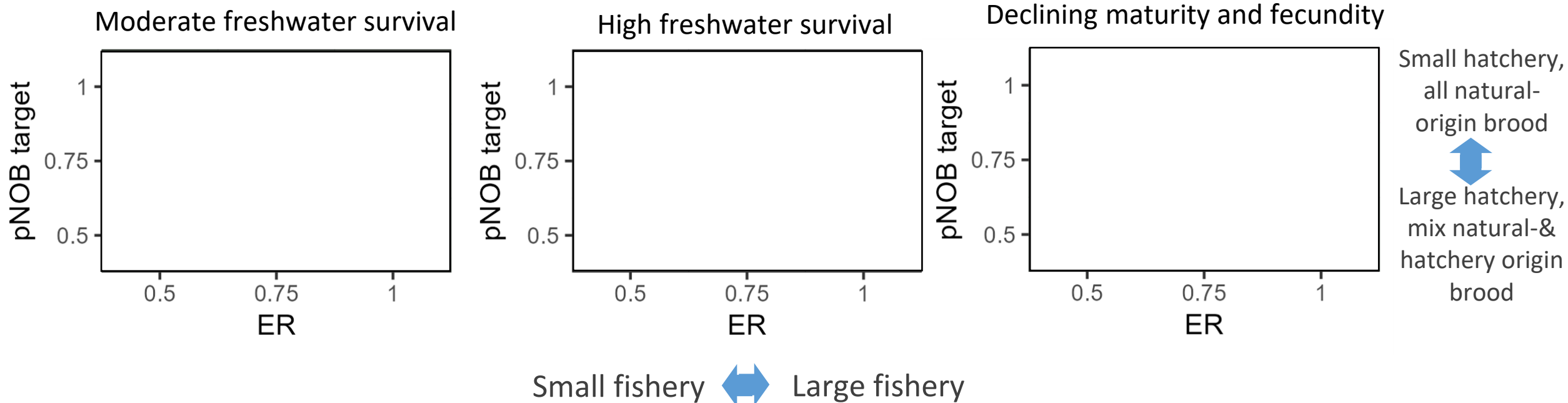
In-river ER = 25%
 $pNOB_{target} = 0.5$

10% freshwater
survival (low)



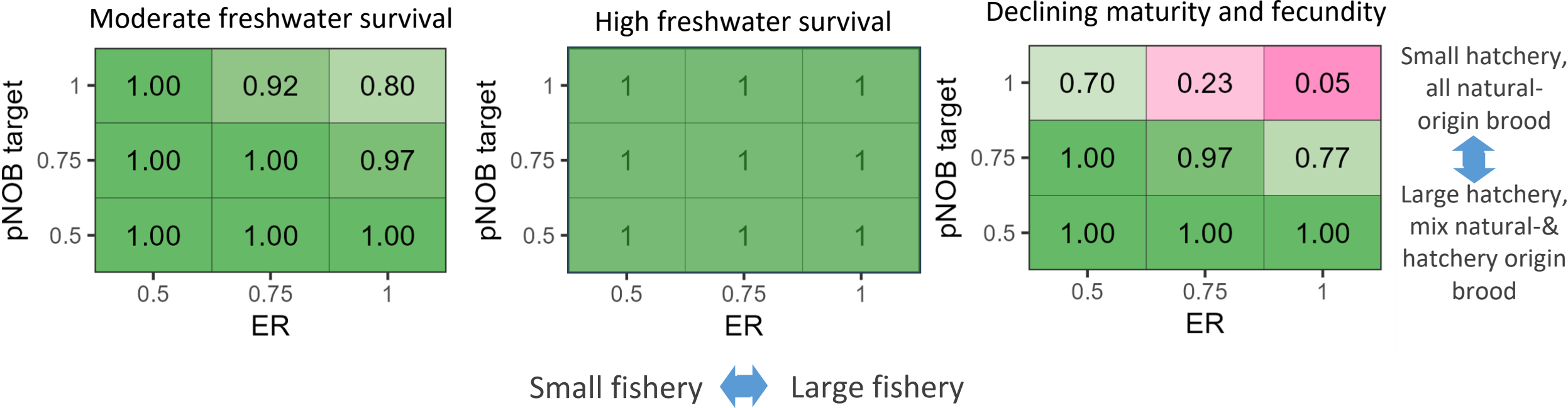
(1) Can the population rebuild above targets under range of harvest and hatchery management options?

Probability of achieving rebuilding target



(1) Can the population rebuild above targets under range of harvest and hatchery management options?

Probability of achieving rebuilding target



Under moderate to high survival, population can be maintained above target, but not under declines in maturity and fecundity with high pNOB targets

(2) Can the population maintain 'wildness' under a range of harvest and hatchery management options?

Degree of 'wildness' (objective = PNI >0.5)

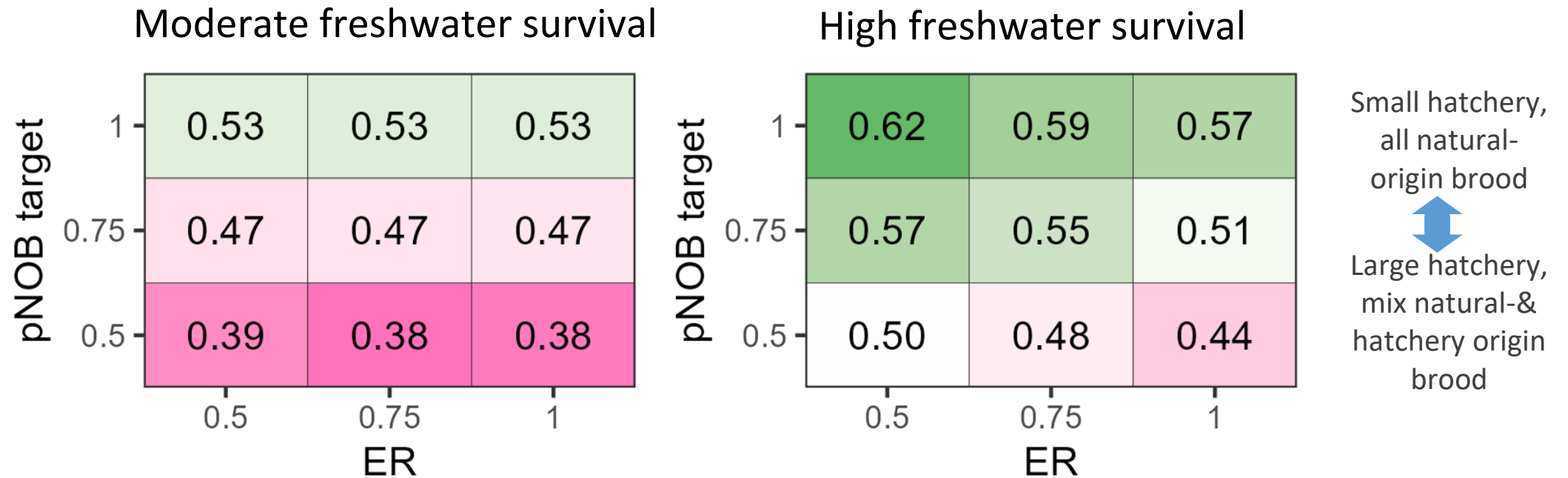
Moderate freshwater survival



Wildness objective is only be met under high pNOB targets under moderate freshwater survival

(2) Can the population maintain 'wildness' under a range of harvest and hatchery management options?

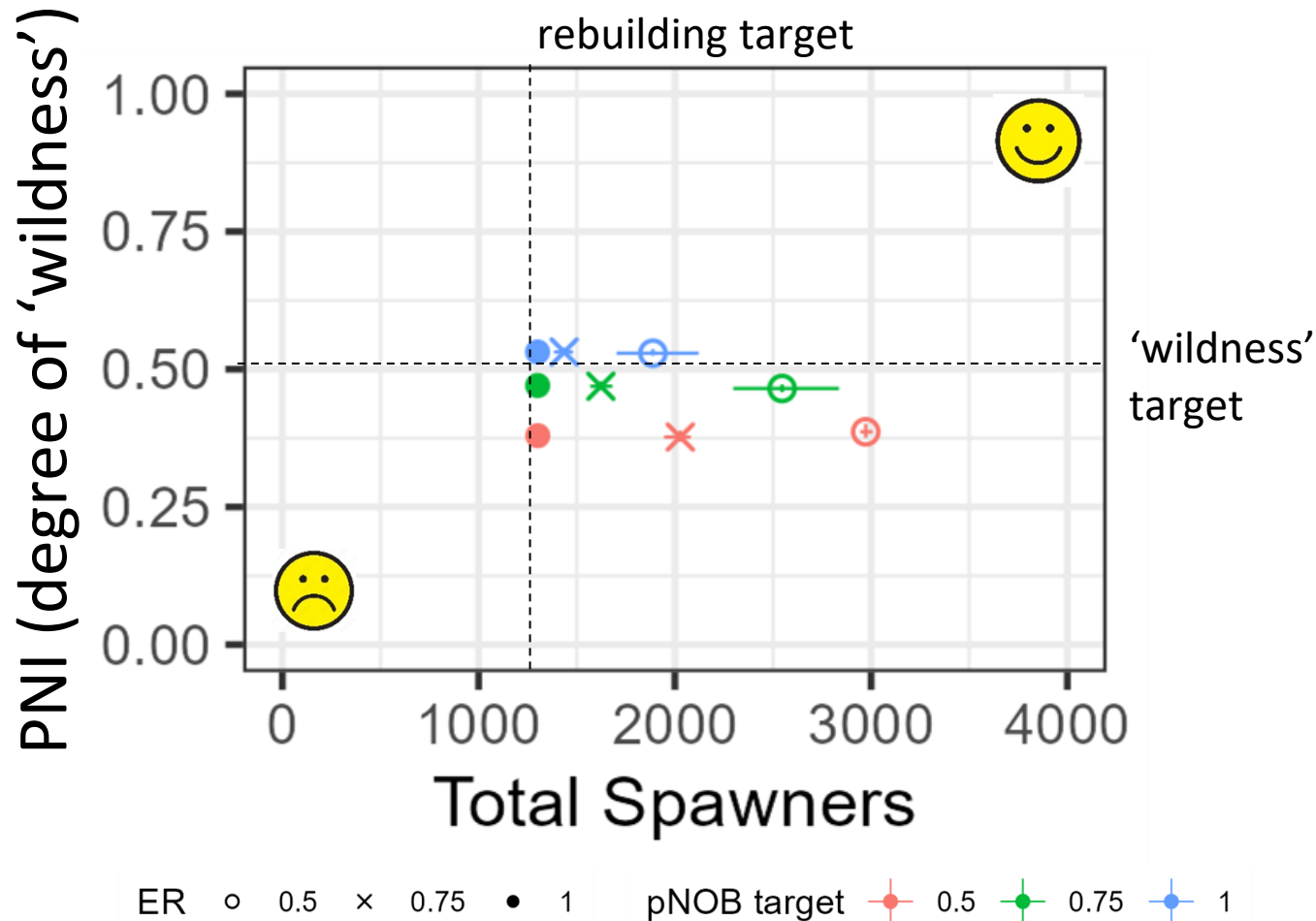
Degree of 'wildness' (objective = PNI >0.5)



Wildness objective is only be met under high pNOB targets under moderate freshwater survival

Under high freshwater survival, wildness can be met at moderate pNOB targets. **Habitat restoration** could support higher freshwater survival to mitigate impacts of poor flow conditions

Trade-offs

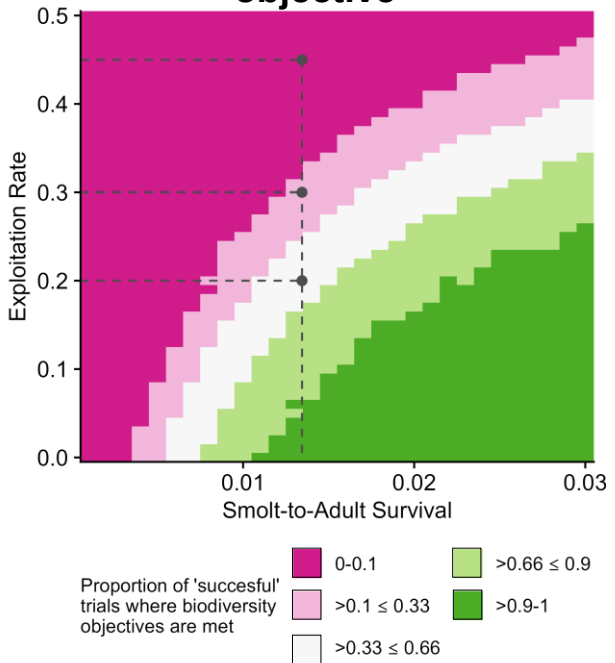


No management option clearly maximizes both.

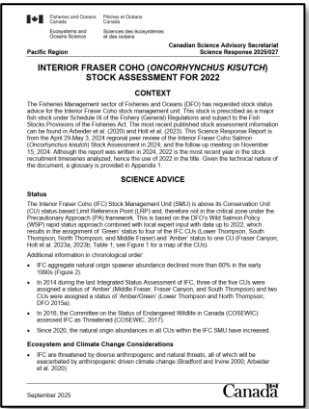
For decision-making, it helps to first prioritize either high PNI or high spawners to select the appropriate management option that aligns with objectives

How can salmonMSE
support FSARs?

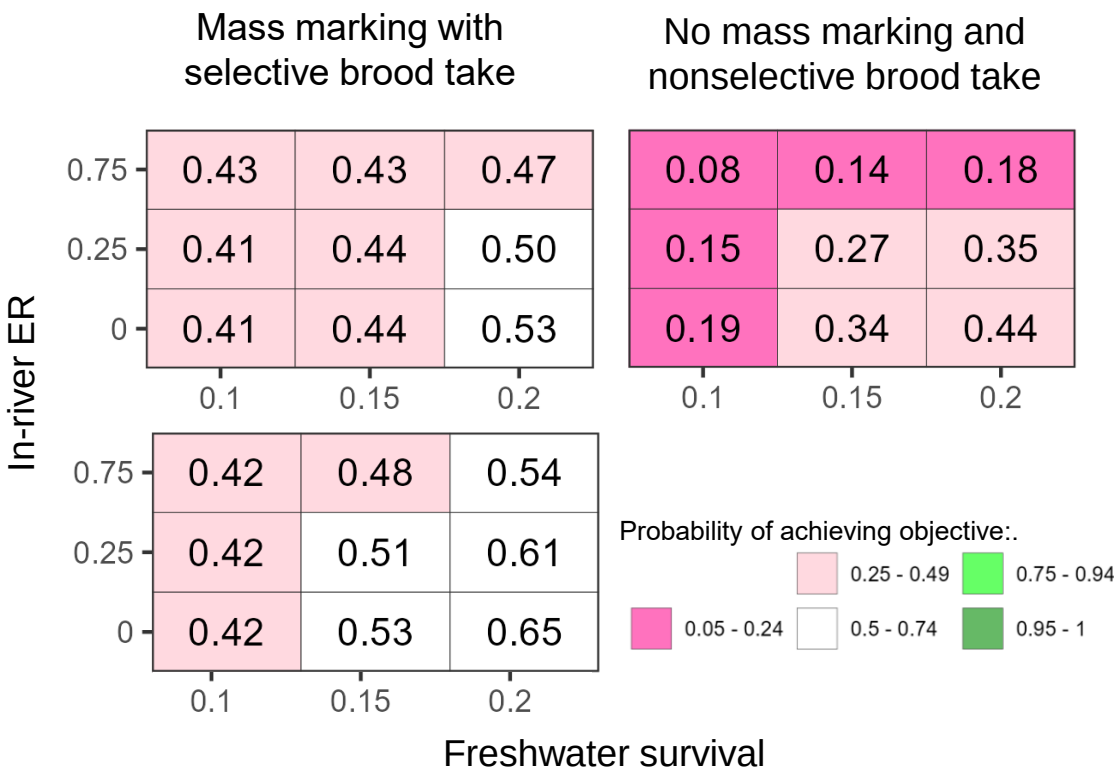
Probability of achieving abundance objective



Interior Fraser River Coho FSAR (DFO 2025)



Probability of achieving PNI objective (PNI> 0.5)



Terminal marine fishery that is **not** mark selective

Terminal marine fishery that is **mark** selective

Saria River Chinook Salmon (salmonMSE)

Developing a visualization app

Standalone Quarto document: https://github.com/Blue-Matter/WCVI_Chinook/blob/main/shiny_quarto/Sarita_shiny.qmd

Sarita visualization app

AUTHOR

Q. Huynh, Blue Matter

PUBLISHED

November 27, 2025

This webpage presents simulation results from the application of candidate management strategies for Sarita Chinook across alternative plausible scenarios. Interactive results are intended to efficiently share comprehensive results with a working group when static figures may be cumbersome to distribute. Figures below show outcomes from user selected scenario, management option, and subset of simulations.

This document is rendered in Quarto and made interactive using Shiny.

Filter results

Select one population scenario (A-E) and one management strategy (1-9).

Scenario

B. High freshwater survival

Management option

(3) ER = 1, pNOB = 0.5

Spaghetti plot

Barplots

All simulations

B. High freshwater survival
(1) ER = 0.5, pNOB = 0.5

Egg (10⁶)

Outmigrating juvenile (10⁵)

Preterminal catch

Mature Return

Terminal catch

In-river Return

Brood

In-river Catch

Natural Spawners

Figure 1: Time series lineplots of state variables for all simulations (grey lines) with annual median in blue.

Sarita visualization app

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Filter results

Select one population scenario (A-E) and one management strategy (1-9).

Scenario

B. High freshwater survival

Management option

(3) ER = 1, pNOB = 0.5

Spaghetti plot

Barplots

Simulation Number

5

B. High freshwater survival
(3) ER = 1, pNOB = 0.5

Egg (10⁶)

Outmigrating juvenile (10⁵)

Preterminal catch

Mature Return

Terminal catch

In-river Return

Brood

In-river Catch

Natural Spawners

Year

0 10 20 30

Hatchery origin

Natural origin

Simulation 5



salmonMSE closed-loop simulation tool to evaluate trade-offs among management actions across various harvest, hatchery and habitat levers

Applicable across a range of data availability

Can be parameterized with available data in conditioning model supplemented by additional information from meta-analyses and expert opinion, e.g., related to stage-specific survival, environmental drivers, and life-history diversity

Open access tool for application across Pacific salmon species and populations



salmonmse.com

Questions



Are there additional examples where salmonMSE may be useful to support decision making?

What value do you see in applying salmonMSE to populations you work with?

Are there ways salmonMSE can be further developed to address management needs? Are you interested in collaborating to extend the tool in new ways?

Are you interested in training in salmonMSE?